

FormFactor, Inc. (NASDAQ: FORM)

The turnaround is real. The price assumes it is permanent.

FormFactor, the largest U.S. maker of wafer probe cards, has staged a sharp recovery: revenue grew 32% year over year in Q2 2026 and non-GAAP gross margin rose from 40.8% in FY2025 to 53.3%. The shares have risen about 285% over the past year to \$149, above the \$139 average analyst target. We see the operational turnaround as genuine, but believe the valuation now discounts peak conditions as the new normal. **Our view is Cautious.**

Investment summary

- **Part of the margin jump will not repeat.** Management attributes about one-third of Q2's sequential margin gain to one-time items (tariff refunds, precious-metal recovery from a plant closure) and puts the normalized level near 51%. Q3 guidance of 54% includes roughly 3 points of tariff refunds, so underlying margin is flat quarter on quarter.
- **Capital intensity is rising again.** 2026 capex is guided to \$140–170M plus \$25–30M of start-up costs for a new Texas facility, after a record \$104M in 2025. From 2021 to 2025 FormFactor already had the highest capex intensity (8.8% of revenue) and the lowest free-cash-flow margin (6.3%) of four U.S. test and packaging peers.
- **The growth engine carries lower margins and a less favorable mix.** HBM is about two-thirds of DRAM probe-card revenue, but the company states DRAM products earn lower margins than Foundry & Logic, and Q3 guidance points to a shift from HBM toward commodity DDR.
- **Competitors are strong.** In 2025 Technoprobe overtook FormFactor's probe-card segment in USD revenue (\$709M vs. \$638M); Micronics Japan earned a 23.6% operating margin versus FormFactor's 7.3%.

What would change our view

Underlying gross margin holding above 51% after one-time items roll off; the Texas facility ramping on schedule and lifting margins by 2028; and free-cash-flow margin moving toward the 15–20% earned by peers despite the investment cycle.

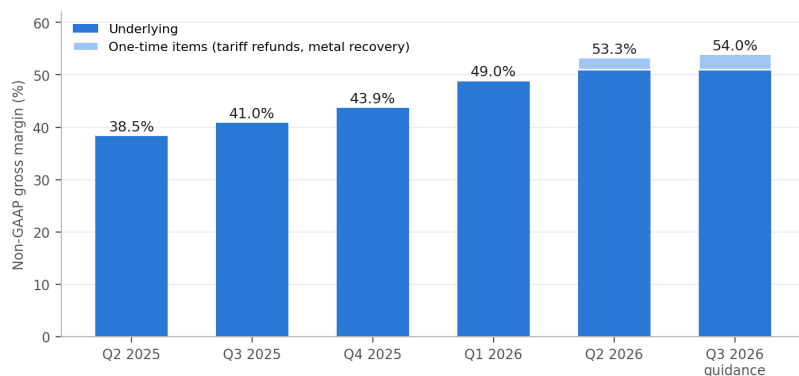


Figure 1. Quarterly non-GAAP gross margin; one-time items per management (Q2 2026 normalized ~51%; Q3 guidance includes ~3 pts of tariff refunds). Source: FormFactor earnings releases, Q2 2026 call; notebook 05.

VIEW: CAUTIOUS

Market data

Price (Oct 2, 2026)	\$149.15
Market cap	\$11.65B
Enterprise value	\$11.34B
Shares outstanding	78.1M
P/E (trailing)	102.9x
P/E (forward)	43.2x
EV / sales	12.6x
52-week change	+285%
Avg. analyst target	\$138.63

Financials (\$M)	FY25	H1 26
Revenue	785.0	484.4
Growth y/y	2.8%	31.9%
Gross margin (GAAP)	39.3%	45.0%
Net income (GAAP)	54.4	76.6
Free cash flow	11.7	83.3
Cash & securities	275.1	345.6
Long-term debt	11.1	11.7

H1 2026 gross margin is GAAP, derived from quarterly figures (38.4% on \$226.1M; 50.7% on \$258.2M). FY25 cash & securities = cash \$103.3M + marketable securities \$171.8M.

Key facts

Products: probe cards (81% of FY25 revenue) and engineering probe systems (19%).

Customers: three customers were 35.3% of Q2 2026 revenue; the largest was 22.9% of FY25 revenue.

Manufacturing: Livermore CA, Beaverton OR; new Farmers Branch TX site ramping from late 2026.

Employees: 2,153 (Dec 2025).

Analyst

Chen-Chih (Cosby) Hung
MS Business Analytics candidate,
University of Illinois
Urbana-Champaign

1. Business and revenue mix

FormFactor sells two kinds of product. **Probe cards** are custom-built interfaces, carrying thousands to over 150,000 contact elements, that connect a tester to every die on a wafer before it is cut and packaged. Each new chip design needs its own card and cards wear out, so demand tracks design activity and wafer output rather than only factory expansion. **Systems** are probe stations and thermal, cryogenic and optical test equipment used mainly in R&D labs.

The mix is shifting toward memory. DRAM revenue nearly tripled from the 2023 trough (\$114M) to \$309M over the twelve months to June 2026. The driver is high-bandwidth memory (HBM) for AI accelerators: HBM stacks 8 to 16 DRAM dies, so every die must be proven good at wafer level before stacking. Foundry & Logic, still the largest market, recovered to a new high on data-center CPU and early custom-ASIC demand. Flash has shrunk to under 3% of revenue.

Revenue by market (\$M)	FY2022	FY2023	FY2024	FY2025	TTM Jun-26
Foundry & Logic	409.2	363.5	381.2	369.9	418.1
DRAM	133.4	113.8	227.4	247.4	309.4
Flash	48.8	20.6	17.4	20.6	19.7
Systems	156.5	165.2	137.6	147.1	155.0
Total	747.9	663.1	763.6	785.0	902.2
DRAM share of revenue	17.8%	17.2%	29.8%	31.5%	34.3%

Table 1. Sums of markets reconcile to SEC-reported revenue for FY2022–FY2025. Sources: FormFactor 10-K (FY2023, FY2025); TTM from stockanalysis.com.

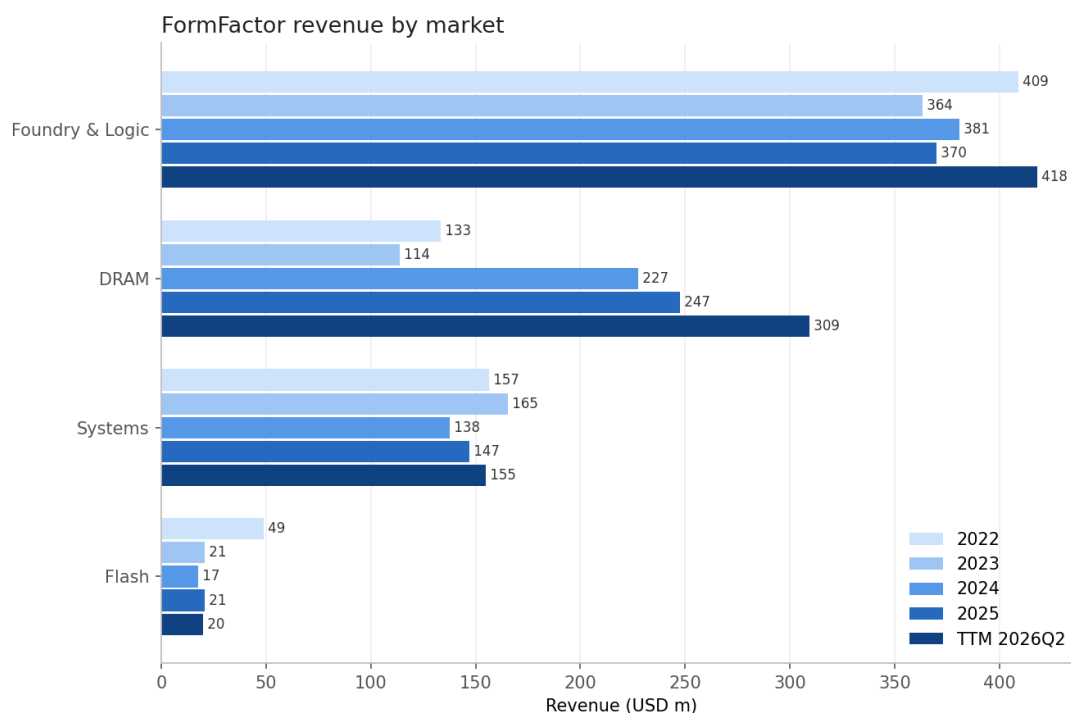


Figure 2. FormFactor revenue by market, FY2022 to trailing twelve months to June 2026.

Downturn behavior. In FY2023 Flash fell 58%, DRAM 15% and Foundry & Logic 11%, while Systems grew 5.5% (about 13% excluding the Metrology business sold in Q4 2023). Revenue growth volatility from 2017 to 2025 was the lowest among five U.S.-listed test and packaging peers, and FormFactor's 2023 decline (–11.3%) was the shallowest of the group.

2. Margins and capital intensity

Through FY2025, revenue growth did not reach the gross margin line. Probe-card gross margin stayed near 40% even as segment revenue rose from \$498M (2023) to \$638M (2025). Management attributed the FY2025 dip to higher manufacturing costs, including tariffs, and noted that *"in general, our DRAM products have lower margins than our Foundry & Logic products."* Systems margin fell about ten points after the sale of the higher-margin Metrology (FRT) business.

Segment gross margin (GAAP)	FY2022	FY2023	FY2024	FY2025
Probe Cards	39.8%	37.2%	41.4%	40.5%
Systems	51.7%	51.3%	43.2%	41.8%
Company total	39.6%	39.0%	40.3%	39.3%

Table 2. Source: FormFactor 10-K (FY2023, FY2025), gross profit and gross margin by segment.

Reported earnings need adjusting. FY2023 operating income of \$82.8M included a \$73.0M gain on the FRT sale; the underlying figure was about \$9.8M. FY2018 net income included a roughly \$70M tax benefit. Adjusted for the FRT gain, the three-year average operating margin halved from 11.1% (2021) to 5.7% (2025).

Operating margin	2021	2022	2023	2024	2025
Reported	12.7%	7.3%	12.5%	8.5%	7.3%
Excluding \$73.0M FRT gain	12.7%	7.3%	1.5%	8.5%	7.3%
3-year average, adjusted	11.1%	10.7%	7.2%	5.8%	5.7%

Table 3. Source: SEC XBRL data; FRT gain per Q4 2023 earnings release.

2026 changed the margin picture. Non-GAAP gross margin reached 49.0% in Q1 and 53.3% in Q2. Management splits the Q2 sequential gain roughly equally between durable cost reductions (including closing the Baldwin Park site), higher volume, and non-recurring items, and puts the normalized level near 51%.

Capital intensity is the persistent weakness. Over 2021–2025 FormFactor spent 8.8% of revenue on capex, about 1.5x Teradyne and more than 3x Onto, Kulicke & Soffa and Cohu, leaving the lowest free-cash-flow margin in the group. Three-year free cash flow fell from \$253M (2020–22) to \$99M (2023–25). The Texas build-out lifts 2026 capex to \$140–170M, partly offset by a \$24.2M state grant; management expects the site to be margin-accretive once it reaches initial target capacity in early 2028.

2021–2025 average

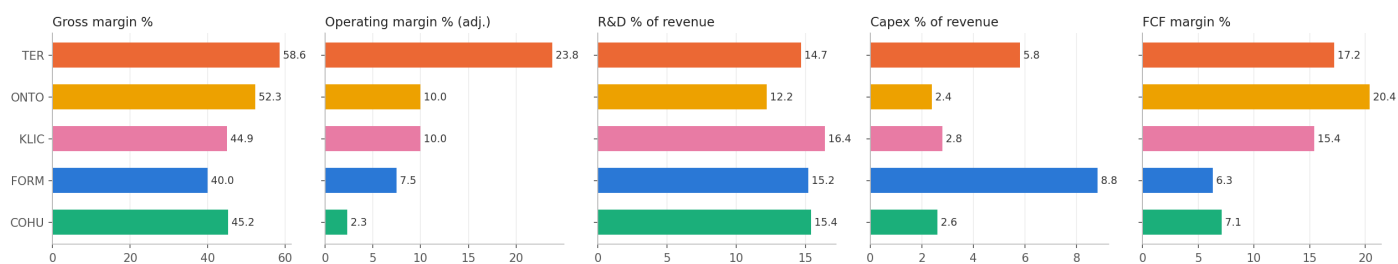


Figure 3. 2021–2025 averages for FormFactor and four U.S.-listed test and packaging peers; FormFactor operating margin excludes the FRT gain. Cohu's cost of revenue excludes D&A from 2021, slightly overstating its gross margin. Source: SEC XBRL data.

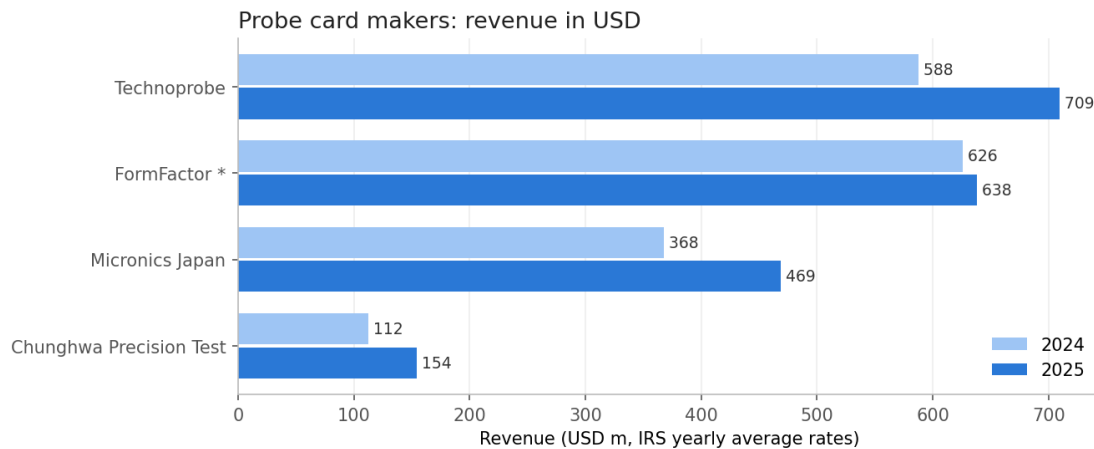
3. Competitive position

The probe-card market is concentrated: an estimated \$2.8B in 2025, with the top three suppliers holding roughly 60%.

FormFactor's direct competitors are not SEC filers, so their results were collected from company reports and converted to USD at IRS yearly average rates.

Probe-card makers, FY2025	Revenue (USD m)	Growth (local)	Net margin	Profitability metric
Technoprobe (Italy)	709.3	+15.7%	15.7%	EBITDA margin 32.1%
FormFactor (U.S.)*	637.9	+1.9%	6.9%	Operating margin 7.3%
Micronics Japan	469.0	+26.1%	17.2%	Operating margin 23.6%
Chunghwa Precision Test (Taiwan)	154.2	+33.3%	n/a	n/a

Table 4. *FormFactor revenue is the Probe Cards segment; its net margin is company-wide. Competitors' figures are total company revenue; Technoprobe includes the Device Interface Solutions business acquired from Teradyne. Chunghwa's growth is based on an estimated 2024 figure. Sources: company reports; IRS exchange rates.



* FormFactor = Probe Cards segment only; others = total company revenue

Figure 4. Probe-card makers' revenue in USD, 2024 and 2025.

2025: share loss. FormFactor's share of the four listed makers' revenue fell from 37.0% to 32.4% as Technoprobe moved into first place; its estimated global share is about 23%. The profitability gap with Micronics Japan and Technoprobe shows that low margins were company-specific rather than structural to probe cards.

2026: partial recovery. H1 2026 revenue grew 31.9%, two customers adopted FormFactor's SmartMatrix full-wafer contactor for HBM4 testing, and the company holds strong positions at two of the three major HBM makers. Competitors' 2026 results have not been incorporated, so whether FormFactor is regaining share remains open. Management describes its share at a large fabless CPU customer as low single digits, which is both an opportunity and a sign of where rivals are entrenched.

Supply chain. Key inputs (ceramic and organic substrates, complex PCBs, contact elements) are sole- or limited-source and bought on purchase orders. Cards are built in the U.S. while 81% of revenue is international, which exposed FormFactor to tariffs in 2025. China fell from 14% to 7% of revenue in 2025 under export restrictions.

4. Valuation

At \$149, FormFactor trades at 43x forward earnings and 12.6x EV/sales, above the \$139 average analyst target. Because revenue is cyclical and recent margins include one-time items, we value the company on mid-cycle earnings rather than the latest quarter, and test what the share price requires.

Illustrative 2027 scenario	Bear	Base	Bull
Revenue (\$M)	1,000	1,200	1,400
Operating margin	18%	25%	30%
EPS (18% tax, 78.1M shares)	\$1.89	\$3.15	\$4.41
Value at 30x–35x EPS	\$57–66	\$94–110	\$132–154
P/E implied by \$149.15	79x	47x	34x

Table 5. Illustrative scenarios, not a price target. 2026 revenue is tracking near \$1.0B (H1 \$484M plus Q3 guidance of \$270M). Q2 2026 non-GAAP operating margin was 27.9%, including non-recurring gross-margin items. Interest income and share count changes are excluded.

Under these assumptions the current price is consistent only with the bull case: revenue of about \$1.4B in 2027 at a 30% operating margin, still valued at roughly 34x earnings. That outcome requires HBM and custom-ASIC demand to keep growing, the Texas facility to ramp smoothly, and normalized gross margin to rise further from about 51% while capex peaks. A full DCF with explicit capex and working-capital assumptions is the next step.

5. Risks to our view

- **Upside: HBM4 and AI demand outrun industry capacity.** If memory makers stay supply-constrained, FormFactor could sustain pricing and utilization, making 51%+ gross margin the floor rather than the ceiling.
- **Upside: new growth vectors.** Co-packaged optics (Keystone Photonics acquisition) and custom ASICs could broaden the customer base beyond a few memory makers.
- **Downside: cycle and mix.** A DDR pricing reversal or HBM digestion would hit the fastest-growing, lower-margin market first.
- **Downside: execution and concentration.** Start-up costs at Texas, tariff policy, and reliance on three customers for about 35% of revenue.

6. Catalysts and what to watch

- Q3 2026 results: underlying gross margin excluding tariff refunds versus the ~51% normalized level.
- Texas (Farmers Branch) ramp from late 2026; pre-production costs of \$25–30M in 2026.
- Technoprobe and Micronics Japan 2026 results, to test whether FormFactor is regaining probe-card share.
- Free-cash-flow conversion through the 2026–27 capex peak.

Appendix: methodology and sources

Financial statements for FormFactor, Teradyne, Cohu, Onto Innovation and Kulicke & Soffa were extracted from the SEC EDGAR XBRL API, cleaned in Python (10-K filtering, fiscal-year alignment, restatement handling) and analyzed in SQLite; FormFactor's 2026 quarters were derived from 10-Q year-to-date figures and reconciled to reported results. Segment, market and competitor data were collected manually from FormFactor 10-K/10-Q filings and earnings releases, Technoprobe and Micronics Japan annual results, Chunghwa Precision Test revenue reports, IRS yearly average exchange rates and Mordor Intelligence (market size). Every manually collected value is stored with its source and URL. Market data from stockanalysis.com as of October 2, 2026. Code and data: github.com/ChenChih-Hung/formfactor-investment-analysis. Code was developed with AI assistance; the analysis and conclusions are the author's.